School of Computer Science and Engineering

Mid Term Examination - Nov 2025

B.Tech in Computer Science and Engineering

Duration : 90 Minutes
Max Marks : 50

Sem V - R1UC501T - Theory of Computation

General Instructions
Answer to the specific question asked
Draw neat, labelled diagrams wherever necessary
Approved data hand books are allowed subject to verification by the Invigilator

1) Define ε–NFA (Epsilon NFA) and explain how it differs from NFA. K1 **(3)**

2) Design an FA which accepts the all set of strings over ∑={a,b} ending with 'aab' K1 **(3)**

3) Explain the difference between Mealy and Moore models of finite state machines. K2 **(4)**

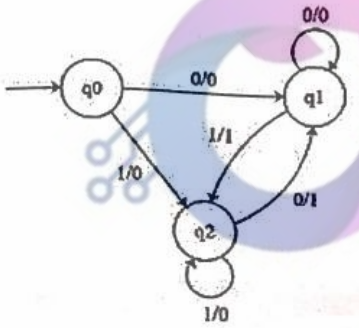
4) Transform the given NDFA to an equivalent DFA. K2 **(4)**

State	*a*	*b*
q_0	q_1, q_3	q_2, q_3
q_1	q_1	q_3
q_2	q_3	q_2
q_3	—	—

5) Design A RE for the entire set of strings over ∑={0,1} containing the substring '110' K3 **(5)**

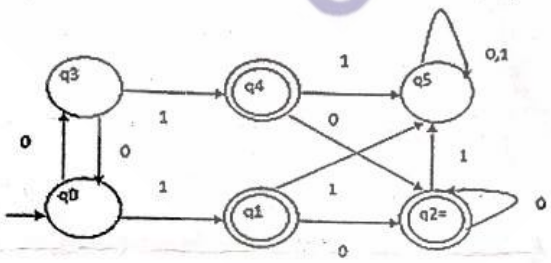
6) Give the formal definition of the NDFA with its tuple. K3 **(5)**

7) Convert the given Mealy Machine to an equivalent Moore Machine. K4 **(8)**



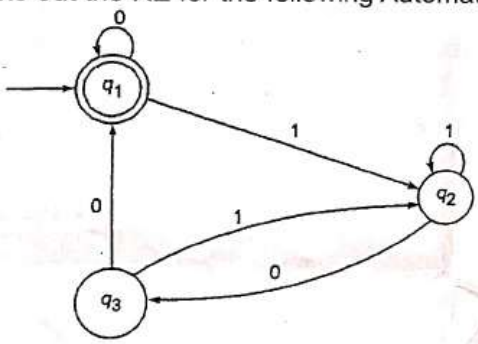
8) For a given RE aba(a+bb)*a Draw the Transition Diagram. K4 **(8)**

9) Find the Minimised Automata for given transition diagram. K5 **(10)**



OR

Take out the RE for the following Automata K5 **(10)**



School of Computing Science and Engineering

Mid Term Examination - Apr 2025

B.TECH in Artificial Intelligence and Data Science

(Applicable for Bachelor of Technology in Computer Science and Engineering, B.TECH in Artificial Intelligence Machine Learning, B.TECH CSE Cloud Computing and Virtualization, B.TECH CSE Computer Network and Cy Security, B.TECH CSE Artificial Intelligence and Machine Learning, B.TECH CSE Data Science, B.TECH CSE D Analytics, B.TECH CSE Full Stack Development, B.TECH CSE Artificial Intelligence, B.TECH CSE with specializat Gaming Technology, B.TECH CSE Business Analytics and Optimization, B.TECH CSE with specialization in Inter Things and Cyber Security Including Block Chain, B.TECH CSE with specialization in Cyber Security, B.TECH Cyber Security and Digital Forensics)

Duration : 90 Minutes
Max Marks : 50

Sem IV - R1UC403B - E2UC401B Operating System

General Instructions
Answer to the specific question asked
Draw neat, labelled diagrams wherever necessary
Approved data hand books are allowed subject to verification by the Invigilator

1) Name the components of an operating system K2(2)
2) Memorize the concept of multiprocessor scheduling K1(3)
3) Analyze the importance of operating system services in modern computing environments. K4(4)
4) Describe the principle of concurrency and its importance. K2(6)
5) Apply the concept of a state transition diagram to explain how a process moves from its creation to termination in a multitasking Operating System. K3(6)

6) Explain the Multilevel Feedback Queue CPU Scheduling. K3(9)
7) Consider following processes with length of CPU burst time in milliseconds. All process arrived in order A, B, C, D all time zero. K4(8)

Process	Burst Time
A	5
B	10
C	2
D	1

(a) Draw Gantt charts illustrating execution of these processes for SJF and round robin (time quantum=1) and FCFS.
(b) Calculate waiting time for each process for each scheduling algorithm.
(c) Calculate average waiting time for each scheduling algorithm.

8) Consider the following set of process with the length of the CPU-burst time in given milliseconds. K4(12)

Draw four Gantt charts illustrating the execution of these processes using FCFS, SJF and RR (quantum=2)scheduling. Also calculate waiting time and turnaround time for each scheduling algorithms.

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5
P5	4	13

School of Computing Science and Engineering

Mid Term Examination - Apr 2025

Bachelor of Technology in Computer Science and Engineering

(Applicable for B.TECH in Artificial Intelligence and Data Science, B.TECH in Artificial Intelligence and Machine Learning, B.TECH CSE Cloud Computing and Virtualization, B.TECH CSE Computer Network and Cyber Security, B.TECH CSE Artificial Intelligence and Machine Learning, B.TECH CSE Data Science, B.TECH CSE Data Analytics, B.TECH CSE Full Stack Development, B.TECH CSE Artificial Intelligence, B.TECH CSE with specialization in Gaming Technology, B.TECH CSE Business Analytics and Optimization, B.TECH CSE with specialization in Internet of Things and Cyber Security Including Block Chain, B.TECH CSE with specialization in Cyber Security, B.TECH CSE Cyber Security and Digital Forensics)

Duration : 90 Minutes
Max Marks : 50

Sem IV - R1UC406B - Data Communication and networking

General Instructions
Answer to the specific question asked
Draw neat, labelled diagrams wherever necessary
Approved data hand books are allowed subject to verification by the Invigilator

1)	Compare the Half Duplex and Full Duplex data flow methods.	K2(2)
2)	Define the concept and application of Circuit-Switched Network.	K1(3)
3)	Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. Calculate channel capacity for this channel.	K4(4)
4)	If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.	K2(6)
5)	With the help of diagram, discuss about phase shift keying (PSK)?	K3(6)
6)	Explain star topology and bus topology with the help of diagram.	K3(9)
7)	Compare the physical structures of LAN, WAN, and MAN networks.	K4(8)
8)	Examine the layers of the OSI model and their functions.	K4(12)

School of Engineering

B.TECH

Mid Term Examination - Nov 2023

Duration : 90 Minutes
Max Marks : 50

Sem I - E2UC102C - Programming for Problem Solving

General Instructions
Answer to the specific question asked
Draw neat, labelled diagrams wherever necessary
Approved data hand books are allowed subject to verification by the Invigilator

1) State the key features of Python? K1 (1)

2) Calculate the output of the following expressions 14/(13 * (12-11)) and 14/13*(12-11). Summarize the output whether they are equal or not. K2 (2)

3) Write a Python program to calculate the sum of Gemetric Progression series: To calculate the geometric sum of a series. Take sum as S, a is the first term in the series. r is the common ratio and n is the number of terms. K3 (3)

4) Write a program in python, and draw the flowchart, to find the numbers in the thousands, hundreds, tens, and ones place for an input number? For example: 256 has 6 ones, 5 tens, and 2 Hundred. K3 (6)

5) Write a Python program count_substring, also write the pseudo code in which the user enters a string and a substring. You have to print the number of times that the substring occurs in the given string. K3 (9)

6) Write a pseudo code and program to print the reverse of a given integer number. K4 (8)

7) Explain the concept of operators in Python, and provide examples of different types of operators used in Python. Discuss the following categories of operators: Arithmetic Operators Comparison Operators Logical Operators For each category, provide a brief description of the operators and give examples to illustrate their usage. Explain when and why each category of operators is commonly used in Python. K5 (15)

8) You are given a list of numbers L=[12, 13,14,15,21]. Write a Python program to perform the following operations: 1.Find the total sum of elements of list. 2.Find the length of list. 3.Find the output of L[:3] K6 (6)

CODE MINDS

School of Computing Science and Engineering

Mid Term Examination - May 2025

Bachelor of Technology in Computer Science and Engineering

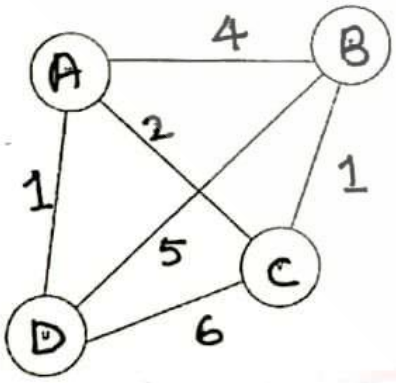
(Applicable for B.TECH in Artificial Intelligence and Data Science, B.TECH in Artificial Intelligence and Machine Learning, B.TECH CSE Cloud Computing and Virtualization, B.TECH CSE Computer Network and Cyber Security, B.TECH CSE Artificial Intelligence and Machine Learning, B.TECH CSE Data Science, B.TECH CSE Data Analytics, B.TECH CSE Full Stack Development, B.TECH CSE Artificial Intelligence, B.TECH CSE with specialization in Gaming Technology, B.TECH CSE Business Analytics and Optimization, B.TECH CSE with specialization in Internet of Things and Cyber Security Including Block Chain, B.TECH CSE with specialization in Cyber Security, B.TECH CSE Cyber Security and Digital Forensics)

Duration : 90 Minutes
Max Marks : 50

Sem IV - R1UC407B - Design and Analysis of Algorithm

General Instructions
Answer to the specific question asked
Draw neat, labelled diagrams wherever necessary
Approved data hand books are allowed subject to verification by the Invigilator

1)	Write down symmetry and transpose symmetry property of asymptotic notations	K2(2)
2)	Define the recurrence relation of recursive function of factorial of number n	K1(3)
3)	What is minimum spanning tree?What is the time complexity of Prim's Algorithm?	K4(4)
4)	Modify a string matching brute force algorithm to improve its efficiency.	K2(6)
5)	Write pseudocode of Krushkal's algorithm	K3(6)
6)	Develop kruskal's algorithm and apply to following graph to find MST.	K3(9)



7)	Write pseudocode of Naïve method of string matching	K4(8)
8)	Calculate time complexity of binary search using sustitution method	K4(12)

CODE MINDS